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Q1. Harshad number

ANSWER

```
n=input().strip()
digit_sum=0
for ch in n:
    digit_sum=digit_sum+int(ch)

num=int(n)
if num % digit_sum==0:
    print("Yes")
else:
    print("No")
```

Q2. GCD

CODE

```
a=int(input("Enter first: "))
b=int(input("\nEnter second: "))
while b!=0:
    temp=a
    a=b
    b=temp%a
print("\nGCD is",a)
```

INPUT

```
15
10
```

OUTPUT

```
Enter first:
Enter second:
GCD is 5
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst,target)
    seen=set()
    pairs=[]
    for num in lst:
        complement=target-num
        if complement in seen:
            pairs.append(complement,num)
        seen.add(num)
    return pairs

data=[int(x) for x in input("Enter numbers: ").split()]
t=int(input("Target: "))
result=find_pairs(data,t)
print("pairs:",result)
```

Q4. Bitwise Operators**ANSWER**

```

step 1: a=0b1101
binary 1101=13
a=13,
data type: int

step 2: 0o25
octal 25=2x8+5=21
b=21
data type: int

step 3: 0x1A
hexadecimal 1A=1x16+10=26
c=26
datatype: int

step 4: result = (a & b) | c
a&b: 13=01101,21=10101->AND=00101=5
5 | 26: 5=00101,26=11010->OR = 11111 =31
result=31,
datatype: int

OUTPUT:
a=13,b=21,c=26, result=31
0b11111 0o37 0x1f
7 62 -32

```

Q5. binary search**CODE**

```

def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[low]==target:
            return mid
        if arr[low]<=arr[mid]:
            if target<=arr[low]and target<arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if target>arr[mid] and target <=arr[high]:
                low=mid+1
            else:
                high=mid-1
    return -1
data=[4,5,6,7,0,1,2]
t=int(input("Target: "))
result=rotated_binary_search(data,t)
print("\nFound at:",result)

```

OUTPUT

```

Target:
Found at: 5

```